

DRY-RUN GUIDE | 05_largest_number.java
Source: java\2nd_belt\05_largest_number.java

CORE BEHAVIOR

Initializes largest to Integer.MIN_VALUE, then replaces it whenever the next input value is greater.

REPRESENTATIVE INPUT / CALL

N=4; values '-9 -3 -12 -6'.

STEP TRACE

First value -9 replaces MIN_VALUE; -3 replaces -9; -12 does not; -6 replaces -3.

EXPECTED OUTPUT / RETURN

'-3'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | /*
002 | Question:
003 | Given N integers, find the largest number in the list.
004 |
005 | Input:
006 | The first line contains an integer N.
007 | The second line contains N integers.
008 |
009 | Output:
010 | Print the largest number.
011 | */
012 |
013 | import java.util.Scanner;
014 |
015 | public class LargestNumber {
016 |     public static void main(String[] args) {
017 |         Scanner sc = new Scanner(System.in);
018 |         int n = sc.nextInt();
019 |
020 |         int largest = Integer.MIN_VALUE;
021 |         for (int i = 0; i < n; i++) {
022 |             int value = sc.nextInt();
023 |             if (value > largest) {
024 |                 largest = value;
025 |             }
026 |         }
027 |
028 |         System.out.println(largest);
029 |         sc.close();
030 |     }
031 | }
032 |
```